

3. [Maximum mark: 6]

A population, P , has a rate of change modelled by $\frac{dP}{dt} = -104000e^{-0.0145t}$, where t is the time measured in years since the **start** of 2022.

At the start of 2022, the population was 6.78×10^6 .

Based on this model, find the predicted population at the start of 2026.

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